Token Distribution & Vesting Simulator

Models monthly circulating supply, supply inflation and supply-driven sell pressure over 48 months. Applied first to a fictional protocol, then to Arbitrum (ARB).

Headline result: in ARB, almost all insider supply arrives in one month (month 12, +77.9% circulating supply), so the same sell assumptions cause a sharper one-month price hit than in a protocol with gradual unlocks.

1. Model design

Each allocation bucket has five parameters:

- **pct**: share of total supply
- **tge**: fraction unlocked at launch (month 0)
- **cliff**: month of the cliff
- **lump**: fraction released at the cliff
- **vest**: months of linear vesting after the cliff

unlocked(t) = tge + lump (if t >= cliff) + (1 - tge - lump) x min(max(t - cliff, 0), vest) / vest

Circulating supply is the sum across buckets. Inflation is the month-over-month % change.

Fictional protocol (1B supply): Team 20%, Investors 18%, Community 22%, Treasury 15%, Liquidity 5%, Future Dev 20%. Team and investors release 25% at the month 12 cliff, then vest linearly.

```
TOTAL = 1_000_000_000
MONTHS = 48

buckets = {
    "Team": dict(pct=20, tge=0.0, cliff=12, lump=0.25, vest=36),
    "Investors": dict(pct=18, tge=0.0, cliff=12, lump=0.25, vest=24),
    "Community": dict(pct=22, tge=0.10, cliff=0, lump=0.0, vest=24),
    "Treasury": dict(pct=15, tge=0.05, cliff=0, lump=0.0, vest=48),
    "Liquidity": dict(pct=5, tge=1.0, cliff=0, lump=0.0, vest=0),
    "Future Dev": dict(pct=20, tge=0.0, cliff=0, lump=0.0, vest=48),
}

print(sum(b["pct"] for b in buckets.values()))
```

```
100
```

```
def unlocked_fraction(t, tge, cliff, lump, vest):
    frac = tge
    if t >= cliff:
        frac += lump
        if vest > 0 and t > cliff:
            months = min(t - cliff, vest)
            frac += (1 - tge - lump) * months / vest
    return frac

team = buckets["Team"]
for t in [0, 11, 12, 13, 24, 48]:
    f = unlocked_fraction(t, team["tge"], team["cliff"], team["lump"], t
    print(t, round(f, 4))
```

```
0 0.0
11 0.0
12 0.25
13 0.2708
24 0.5
48 1.0
```

2. Circulating supply and inflation

Starting float is about 8% of supply. At month 12 the team and investor lumps land together, taking circulating supply from about 248.7M to 359.1M tokens (+44.4% in one month).

```
import pandas as pd

def bucket_curve(b):
    amount = TOTAL * b["pct"] / 100
    return [amount * unlocked_fraction(t, b["tge"], b["cliff"], b["lump"

supply = pd.DataFrame({name: bucket_curve(b) for name, b in buckets.item
supply["Circulating"] = supply.sum(axis=1)
supply["Inflation"] = supply["Circulating"].pct_change() * 100

show = supply[["Circulating", "Inflation"]].iloc[[0, 1, 11, 12, 24, 48]]
print((show / [1e6, 1]).round(2))
```

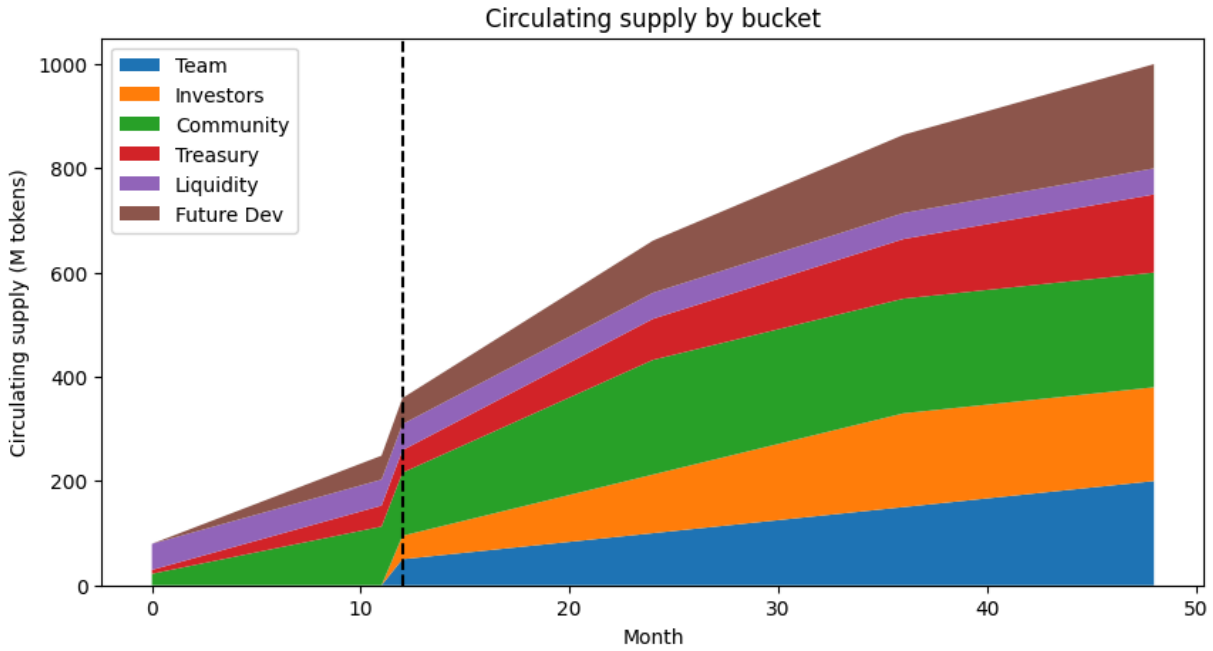
```
    Circulating  Inflation
0         79.50        NaN
1         94.89      19.35
11       248.74       6.59
12       359.12      44.38
```

```
24       661.25       3.96
48      1000.00       1.14
```

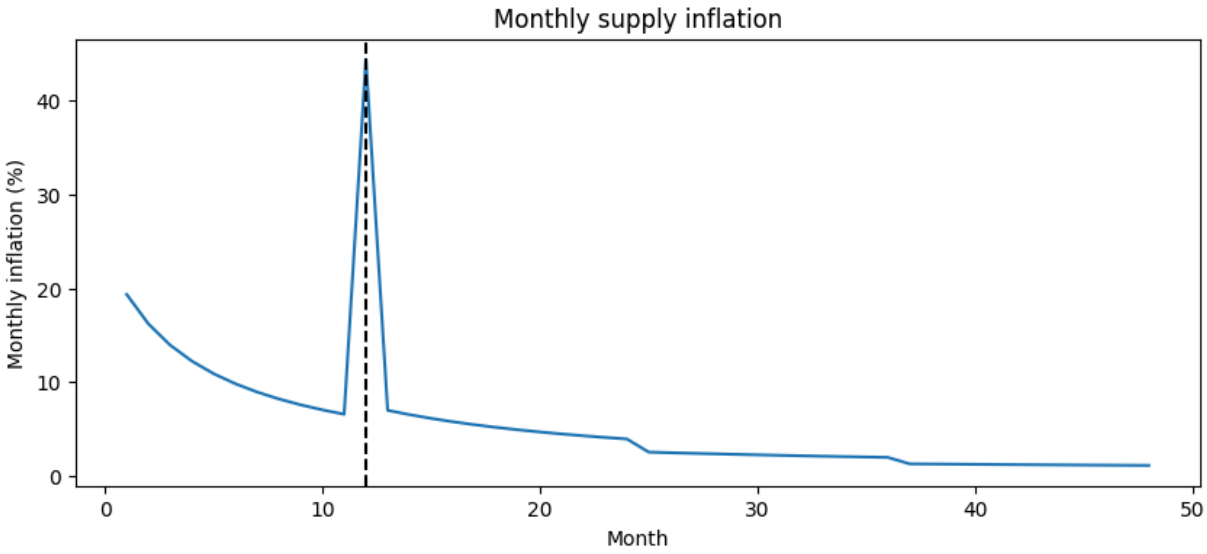
```
import matplotlib.pyplot as plt

months = supply.index
parts = supply.drop(columns=["Circulating", "Inflation"])

plt.figure(figsize=(10, 5))
plt.stackplot(months, parts.T.values / 1e6, labels=parts.columns)
plt.axvline(12, color="black", linestyle="--")
plt.xlabel("Month")
plt.ylabel("Circulating supply (M tokens)")
plt.title("Circulating supply by bucket")
plt.legend(loc="upper left")
plt.show()
```




```
plt.figure(figsize=(10, 4))
plt.plot(months[1:], supply["Inflation"].iloc[1:])
plt.axvline(12, color="black", linestyle="--")
plt.xlabel("Month")
plt.ylabel("Monthly inflation (%)")
plt.title("Monthly supply inflation")
plt.show()
```



3. Sell pressure model

For each month:

1. New unlocks = change in each bucket's unlocked tokens
2. Tokens sold = new unlocks x sell fraction for that bucket
3. Pressure = tokens sold / monthly volume, where volume = previous circulating supply x daily turnover x 30
4. Monthly price impact = -K x pressure, compounded into a price index (start = 100)

Assumptions: TURNOVER = 3% of float traded daily, K = 0.5, demand held constant (this isolates supply effects only).

Sell fractions apply to each month's *new unlocks*. Three scenarios, because true insider behavior is unknown:

	Low	Base	High
Team	10%	20%	50%
Investors	30%	50%	70%
Community	25%	40%	50%
Treasury	0%	5%	10%
Future Dev	5%	10%	20%

```
scenarios = {
    "Low": dict(Team=0.10, Investors=0.30, Community=0.25, Treasury=0.0,
    "Base": dict(Team=0.20, Investors=0.50, Community=0.40, Treasury=0.0
    "High": dict(Team=0.50, Investors=0.70, Community=0.50, Treasury=0.1
}

TURNOVER = 0.03
K = 0.5
```

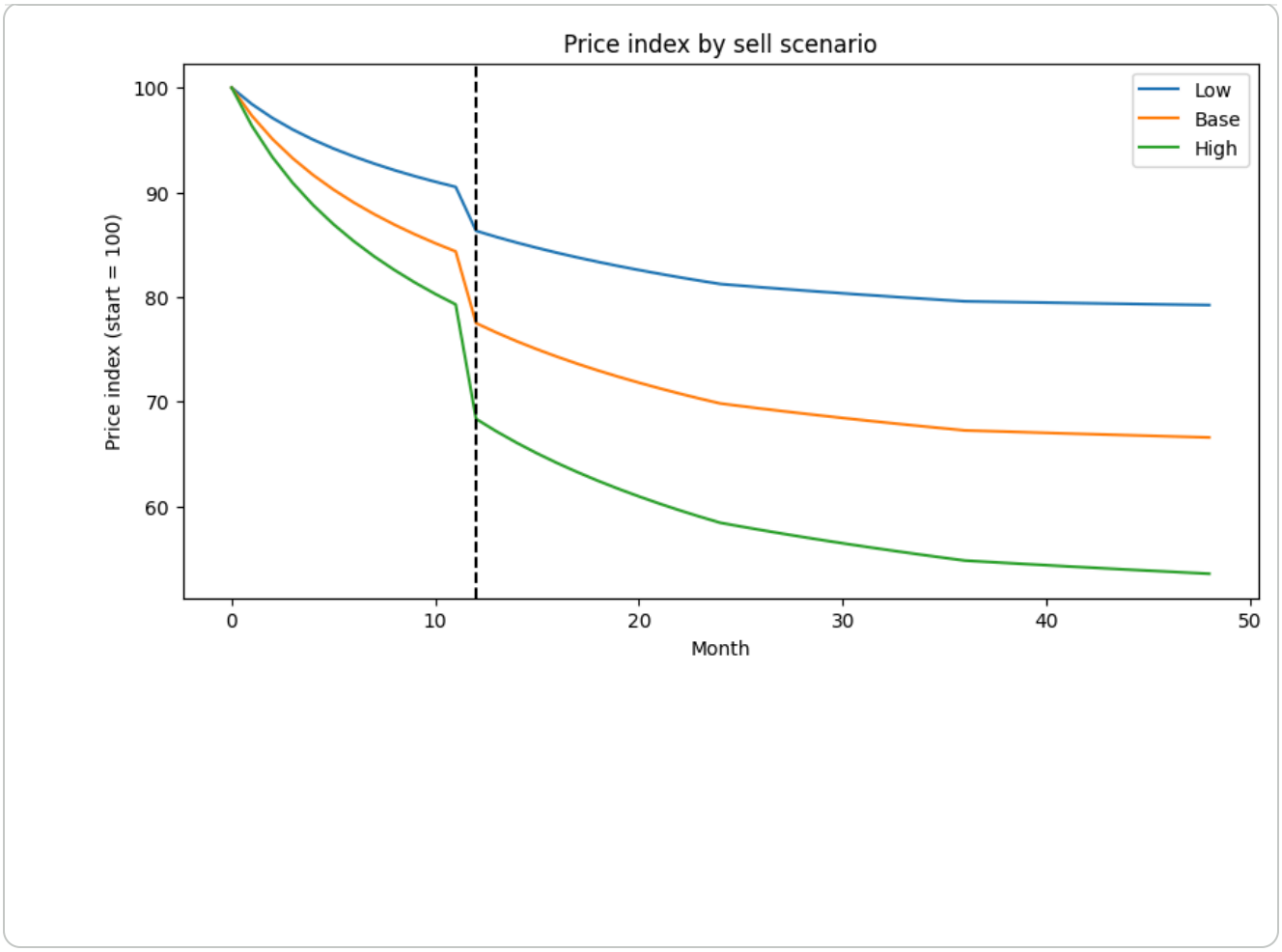
```
parts = supply[list(buckets.keys())]
new = parts.diff().fillna(0)
volume = supply["Circulating"].shift(1) * TURNOVER * 30
prices = pd.DataFrame()

for name, sf in scenarios.items():
    sold = (new * pd.Series(sf)).sum(axis=1)
    impact = -K * (sold / volume).fillna(0)
    prices[name] = 100 * (1 + impact).cumprod()

print(prices.iloc[[0, 11, 12, 24, 48]].round(1))
```

```
      Low   Base   High
0   100.0  100.0  100.0
11   90.5   84.3   79.3
12   86.3   77.5   68.3
24   81.2   69.8   58.4
48   79.2   66.6   53.5
```

```
prices.plot(figsize=(10, 5), title="Price index by sell scenario")
plt.axvline(12, color="black", linestyle="--")
plt.xlabel("Month")
plt.ylabel("Price index (start = 100)")
plt.show()
```



4. Results: fictional protocol

Month 12 price index: Low = 86.3, Base = 77.5, High = 68.3.

Price pressure builds gradually before the cliff (Community unlocks monthly), then the cliff adds a second hit.

5. Applying the model to Arbitrum (ARB)

Data: allocation and vesting from Token Terminal's ARB vesting page, cross-checked against Tokenomist. These are compiled from docs and forum posts, not on-chain data.

Bucket	% of 10B	Treatment
Team + advisors	26.94	1-year cliff, 25% lump, then 36 monthly releases
Investors	17.53	Same schedule as team
Airdrops (users + DAOs)	12.75	100% at launch
Arbitrum Foundation	7.50	Linear over 48 months
DAO treasury	35.28	**Excluded** from circulating supply

Modeling choices:

- The treasury is "unlocked" but spent only by governance votes, so counting it as circulating would misstate the float.
- The 25% lump is derived: 4-year monthly vesting with a 1-year cliff means 12 of 48 months accrue at the cliff.
- Foundation vesting terms come from a secondary source and should be verified against Arbitrum's docs.

```
ARB_TOTAL = 10_000_000_000

arb_buckets = {
    "Team": dict(pct=26.94, tge=0.0, cliff=12, lump=0.25, vest=36),
        "Investors": dict(pct=17.53, tge=0.0, cliff=12, lump=0.25, vest=
            "Airdrops": dict(pct=12.75, tge=1.0, cliff=0, lump=0.0, vest
                "Foundation": dict(pct=7.5, tge=0.0, cliff=0, lump=0.0,
                }
print(sum(b["pct"] for b in arb_buckets.values()))
```

```
64.72
```

```
def arb_curve(b):
    amount = ARB_TOTAL * b["pct"] / 100
    return [amount * unlocked_fraction(t, b["tge"], b["cliff"], b["lump"

arb = pd.DataFrame({name: arb_curve(b) for name, b in arb_buckets.items(
arb["Circulating"] = arb.sum(axis=1)
arb["Inflation"] = arb["Circulating"].pct_change() * 100

show = arb[["Circulating", "Inflation"]].iloc[[0, 11, 12, 13, 24, 48]]
print((show / [1e6, 1]).round(2))
```

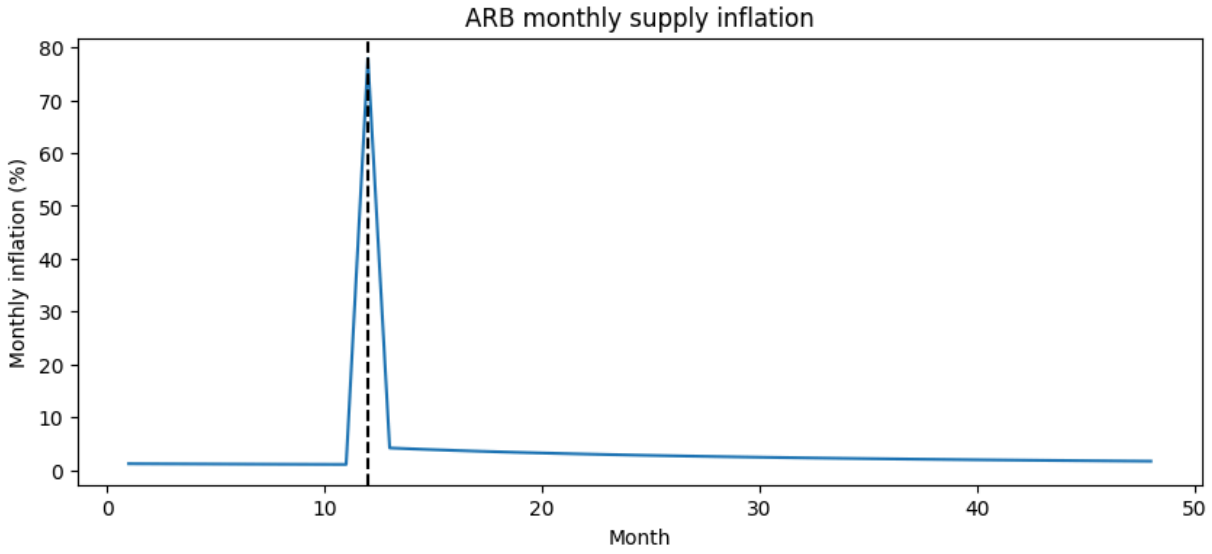
```
    Circulating  Inflation
0       1275.00        NaN
11      1446.88       1.09
12      2574.25      77.92
13      2682.52       4.21
24      3873.50       2.88
48      6472.00       1.70
```

```
arb_low = {"Team": 0.10, "Investors": 0.30, "Airdrops": 0.0, "Foundation
arb_base = {"Team": 0.20, "Investors": 0.50, "Airdrops": 0.0, "Foundatio
arb_high = {"Team": 0.50, "Investors": 0.70, "Airdrops": 0.0, "Foundatio

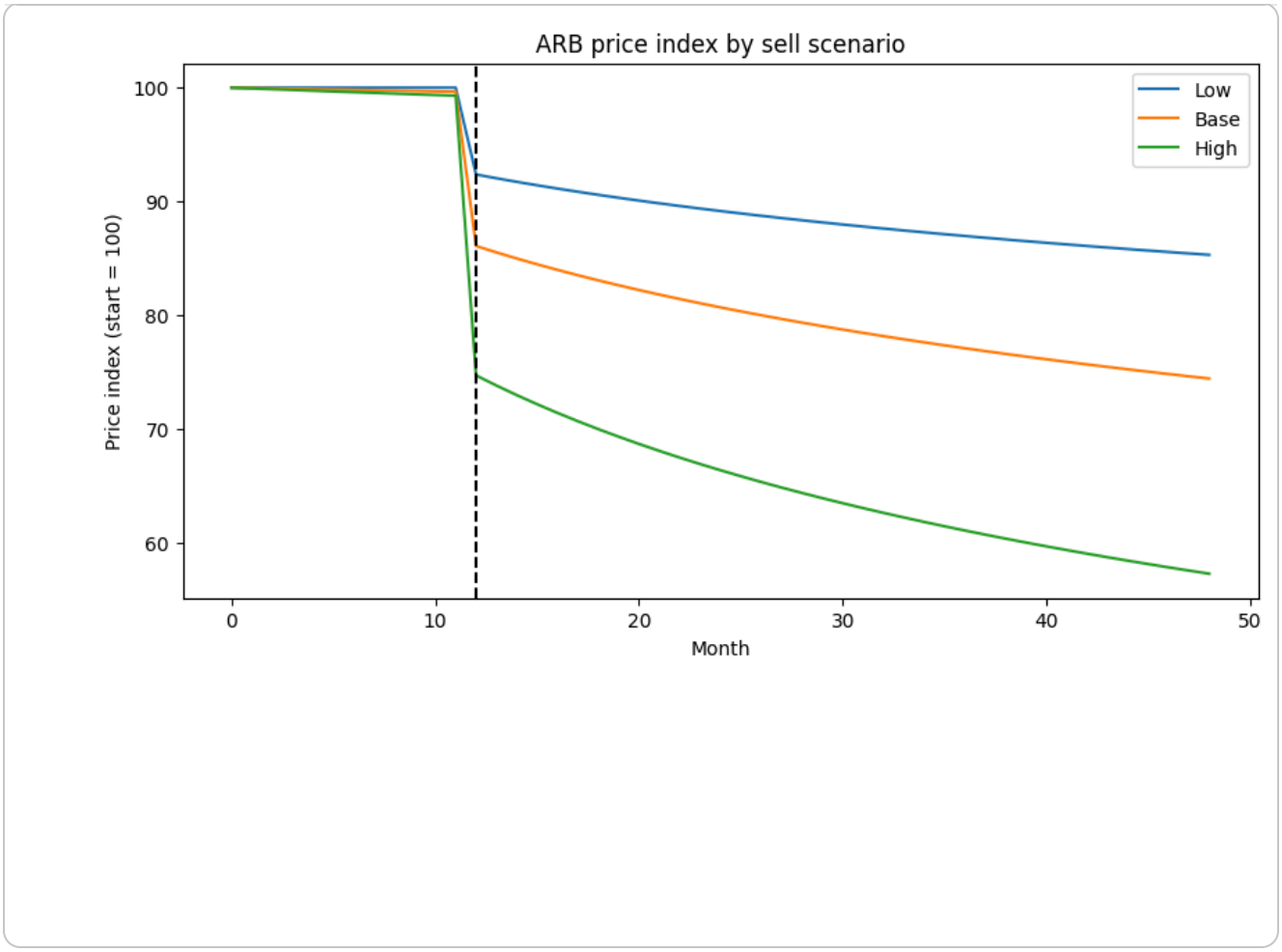
arb_scenarios = {"Low": arb_low, "Base": arb_base, "High": arb_high}
```

```
arb_parts = arb[list(arb_buckets.keys())]
arb_new = arb_parts.diff().fillna(0)
arb_volume = arb["Circulating"].shift(1) * TURNOVER * 30
```

```
plt.figure(figsize=(10, 4))
plt.plot(arb.index[1:], arb["Inflation"].iloc[1:])
plt.axvline(12, color="black", linestyle="--")
plt.xlabel("Month")
plt.ylabel("Monthly inflation (%)")
plt.title("ARB monthly supply inflation")
plt.show()
```




```
arb_prices.plot(figsize=(10, 5), title="ARB price index by sell scenario
plt.axvline(12, color="black", linestyle="--")
plt.xlabel("Month")
plt.ylabel("Price index (start = 100)")
plt.show()
```



6. Results: ARB

Month 12 unlocks about 1.11B tokens (team 673.5M + investors 438.25M) against a float of about 1.45B, so circulating supply rises about 78% in one month.

Month 12 price index: Low ~92, Base ~86, High ~75.

Compared with the fictional protocol, ARB shows almost no pressure before the cliff and takes nearly all the damage in a single month.

7. Case studies

Optimism (OP): 4-year monthly vesting with a 1-year cliff for core contributors (19%) and investors (17%). The first airdrop was about 5% of supply, so the initial float was low. Governance-controlled pools (ecosystem fund, RetroPGF) behave like ARB's treasury: hidden supply that depends on votes.

dYdX (DYDX): about 50% insiders, 45% rewards, 5% DAO (Blockcrunch). Supply grows through continuous trading and liquidity rewards, not just a cliff. The v4 migration

coincided with an unlock that raised circulating supply about 80% overnight.

Pattern	Example	Effect
Cliff concentration	ARB	One-month supply shock
Continuous rewards	DYDX	Steady drip, recipients likely to sell
Low float at launch	OP	Any unlock is large relative to float
Governance-held pools	ARB, OP	Largest uncertainty in supply models

These are structural risk patterns. Actual post-unlock price reactions are not analyzed here.

Sources: Token Terminal (ARB and OP vesting), Tokenomist (ARB, OP, DYDX allocations),